

OSI Reference Model

CEN 223 - Internet Communication

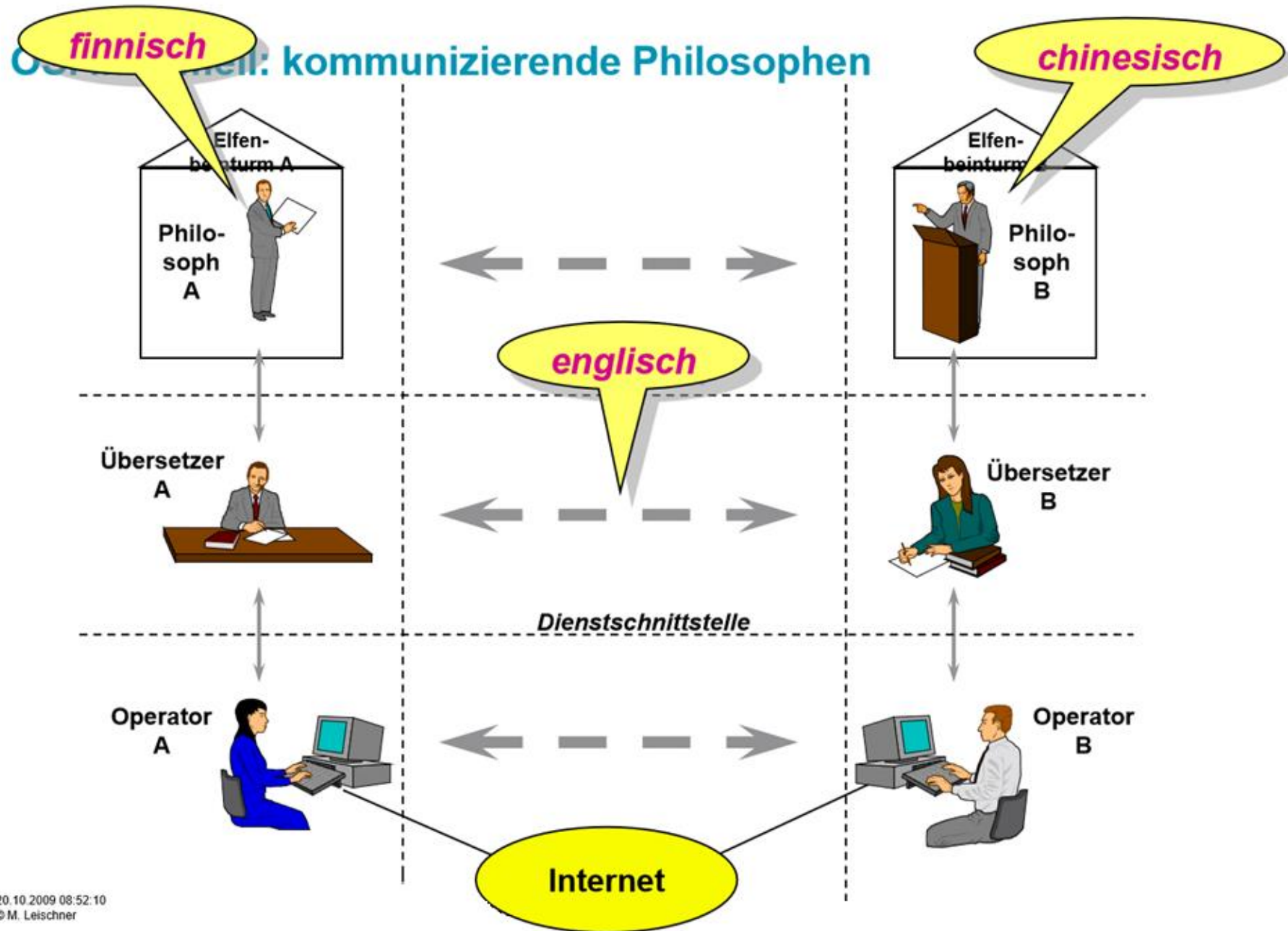
Assoc. Prof. Dr. Fatih ABUT

Çukurova University

Department of Computer Engineering

OSI Reference Model

- ▶ OSI Reference Model - internationally standardised network architecture.
- ▶ OSI = *Open Systems Interconnection*: deals with *open systems*, i.e. systems open for communications with other systems.
- ▶ Specified in ISO 7498.
- ▶ Model has 7 layers.



OSI Features

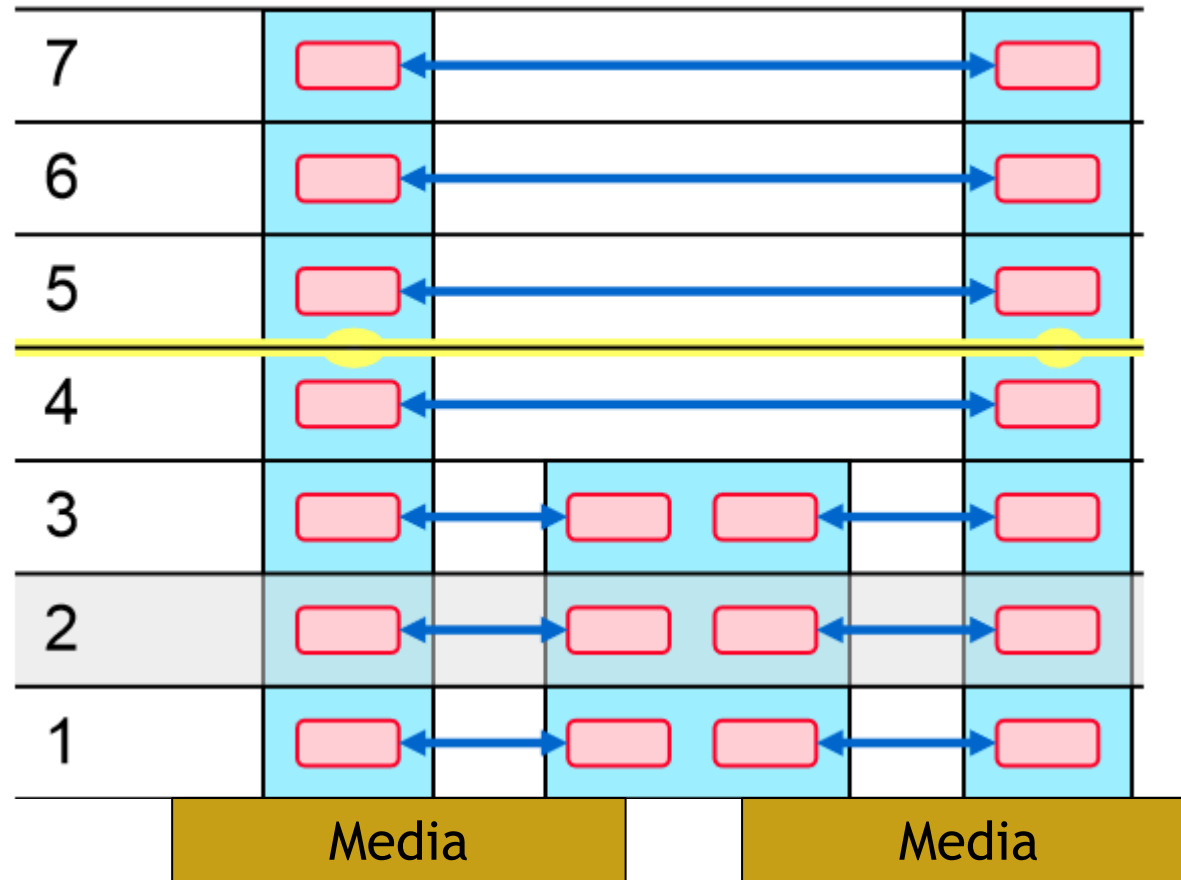
- ▶ Open system standards over the world
- ▶ Rigorously defined structured, hierarchical network model
- ▶ Commonly defined terminology
- ▶ Complete description of the function (not the implementation!)
- ▶ Provide standard test procedures

OSI History

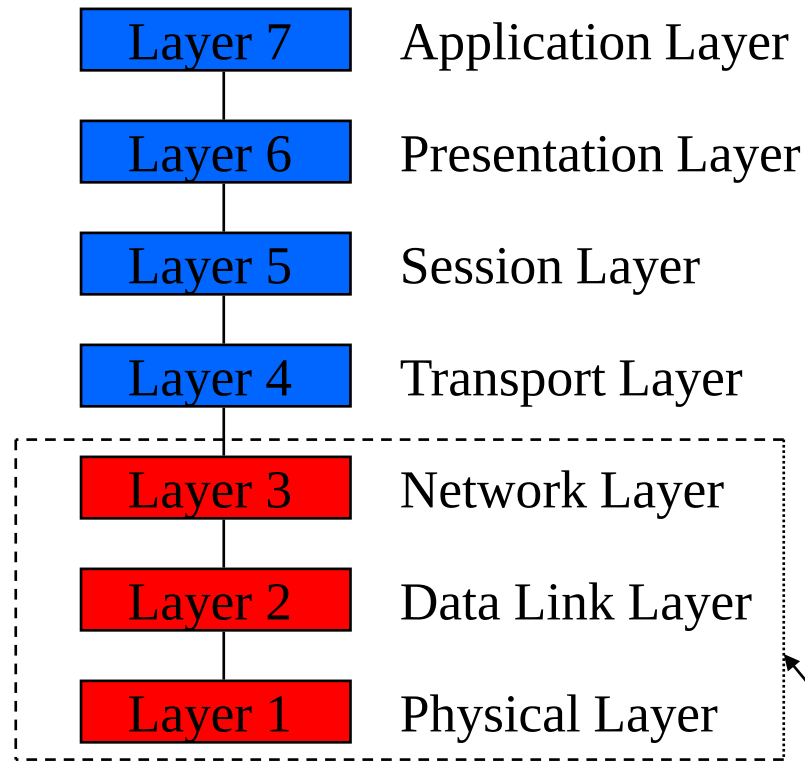
- ▶ In 1978, the International Standard Organization (ISO) began to develop its OSI framework architecture.
- ▶ OSI has two major components: an abstract model of networking, called the Basic Reference Model or seven-layer model, and a set of specific protocols.
- ▶ The concept of a 7 layer model was provided by the work of Charles Bachman, then of Honeywell.

Components of OSI-Model

- Layers
- Systems
- Media
- Services
- Entities
- Protocols



7-Layer OSI Model



► Layers 1-4 relate to communications technology.

► Layers 5-7 relate to user applications.

Communications subnet boundary

Layer 1: Physical Layer

- ▶ Transmits bits from one computer to another
- ▶ Regulates the transmission of a stream of bits over a physical medium.
- ▶ Defines how the cable is attached to the network adapter and what transmission technique is used to send data over the cable. Deals with issues like
 - ▶ The definition of 0 and 1, e.g. how many volts represents a 1, and how long a bit lasts?
 - ▶ Whether the channel is simplex or duplex?
 - ▶ How many pins a connector has, and what the function of each pin is?

Layer 2: Data Link Layer

- Packages raw bits from the Physical layer into frames (logical, structured packets for data).
- Provides reliable transmission of frames
 - It waits for an acknowledgment from the receiving computer.
 - Retransmits frames for which acknowledgement not received
- Access to shared medium

Layer 3: Network Layer

- ▶ Manages addressing/routing of data within the subnet
 - ▶ Addresses messages and translates logical addresses and names into physical addresses.
 - ▶ Determines the route from the source to the destination computer
 - ▶ Manages traffic problems, such as switching, routing, and controlling the congestion of data packets.
- ▶ Routing can be:
 - ▶ Based on static tables
 - ▶ determined at start of each session
 - ▶ Individually determined for each packet, reflecting the current network load.

Layer 4: Transport Layer

- ▶ Manages transmission packets
 - ▶ Repackages long messages when necessary into small packets for transmission
 - ▶ Reassembles packets in correct order to get the original message.
- ▶ Handles error recognition and recovery.
 - ▶ Transport layer at receiving acknowledges packet delivery.
 - ▶ Resends missing packets
- ▶ Flow and congestion control
 - ▶ Avoids congesting the receiver
 - ▶ Avoids congesting the network

Layer 5: Session Layer

- ▶ Allows two applications on different computers to establish, use, and end a session.
 - ▶ e.g. file transfer, remote login
- ▶ Establishes dialog control
 - ▶ Regulates which side transmits, plus when and how long it transmits.
- ▶ Performs *token management* and *synchronization*.

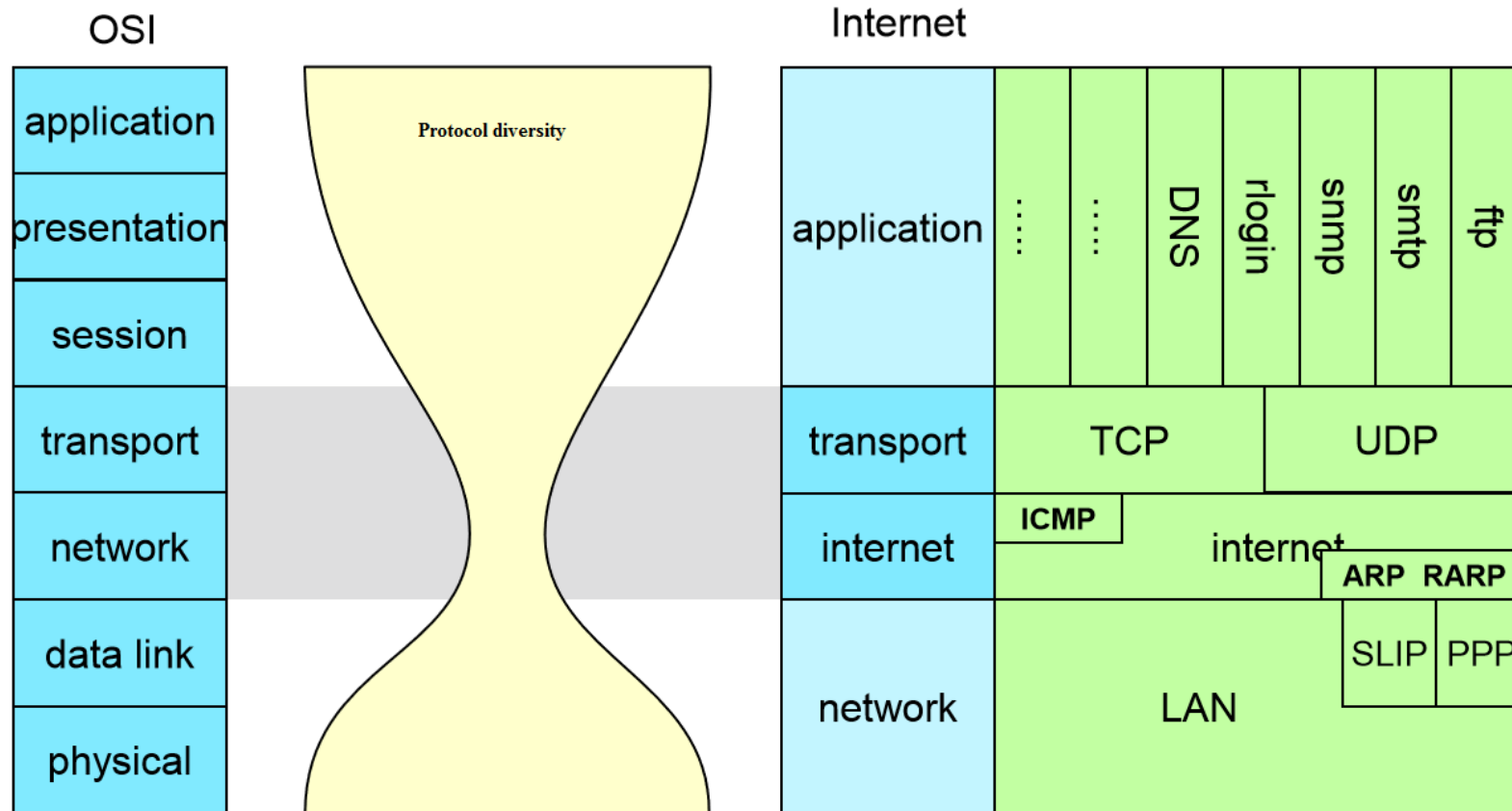
Layer 6: Presentation Layer

- ▶ Related to representation of transmitted data
 - ▶ Translates different data representations from the Application layer into uniform standard format
- ▶ Providing services for secure efficient data transmission
 - ▶ e.g. data encryption, and data compression.

Layer 7: Application Layer

- ▶ Level at which applications access network services.
 - ▶ Represents services that directly support software applications for file transfers, database access, and electronic mail etc.
- ▶ Offers application-specific functionality but does not contain the applications themselves.

Internet Protocols vs. OSI

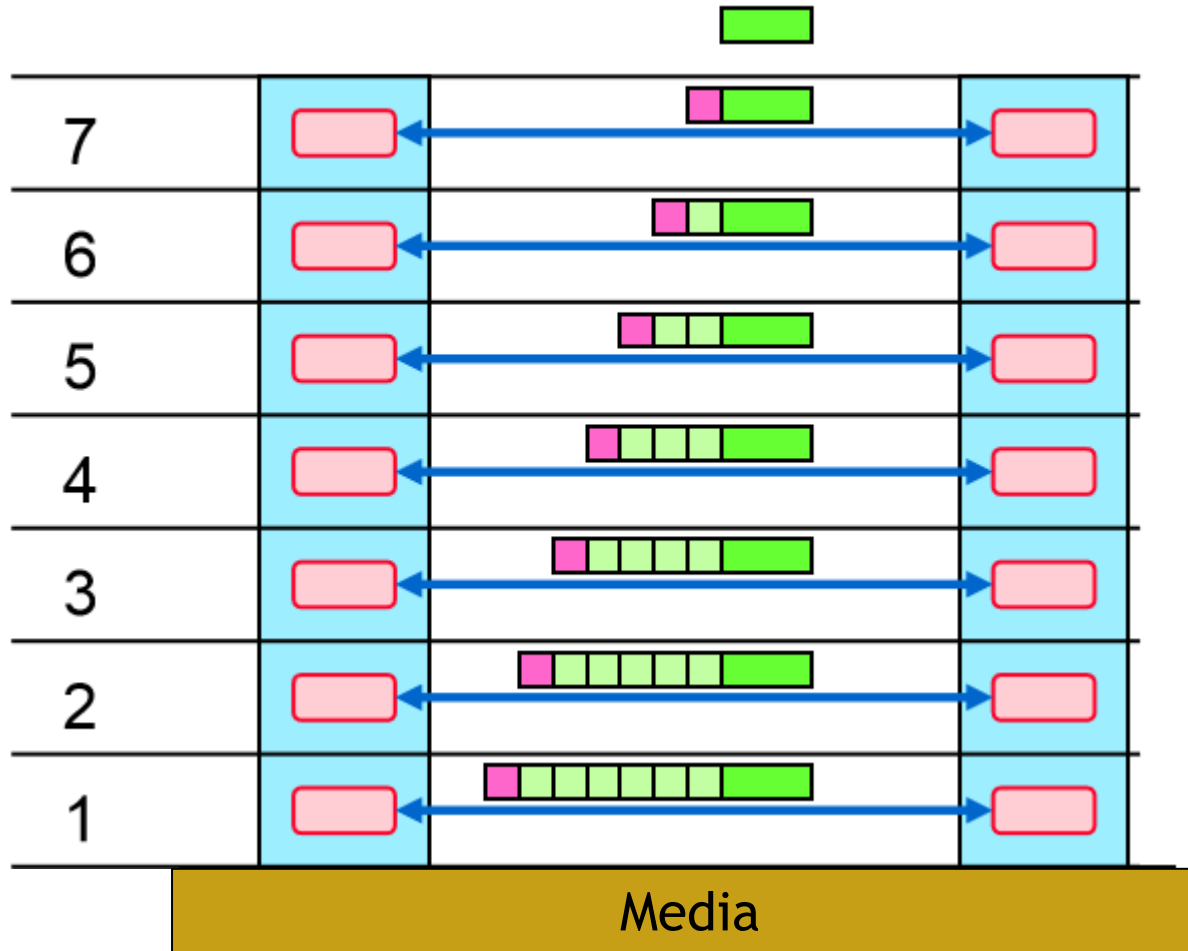


- Explicit presentation and session layers missing in Internet Protocols
- Data Link and Network Layers redesigned

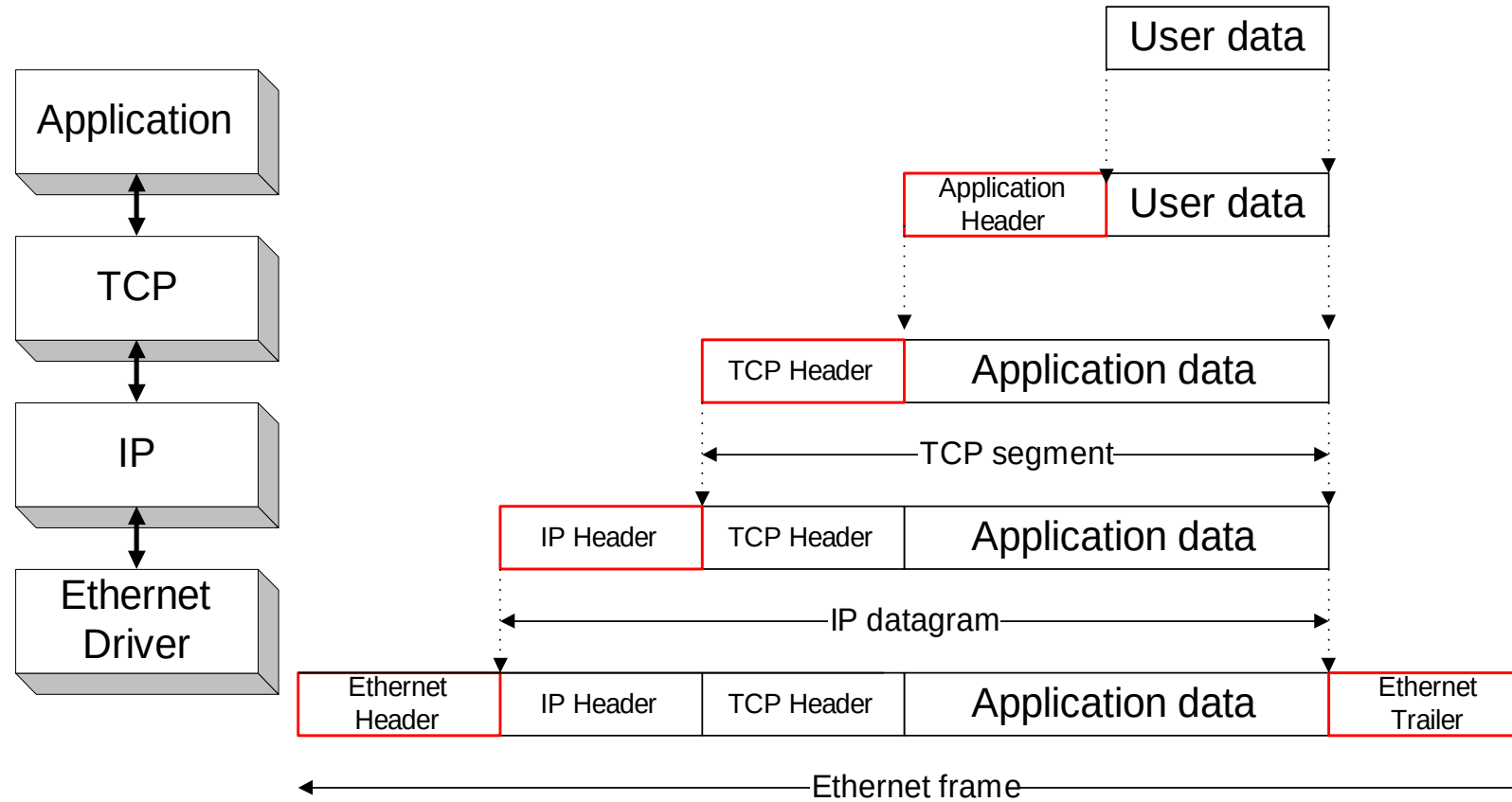
Data Encapsulation in OSI model (1)

- ▶ At each layer in the TCP/IP protocol stack
 - ▶ Outgoing data is packaged and identified for delivery to the layer underneath
- ▶ PDU - Packet Data Unit - the “envelop” information attached to a packet at a particular TCP/IP protocol
 - ▶ e.g. header and trailer
- ▶ Header
 - ▶ PDU’s own particular opening component
 - ▶ Identifies the protocol in use, the sender and intended recipient
 - ▶ The information header is used to assist in any of these tasks: routing of the object, flow control, error detection, error correction, etc.
- ▶ Trailer (or packet trailer)
 - ▶ Provides data integrity checks for the payload

Data Encapsulation in OSI model (2)



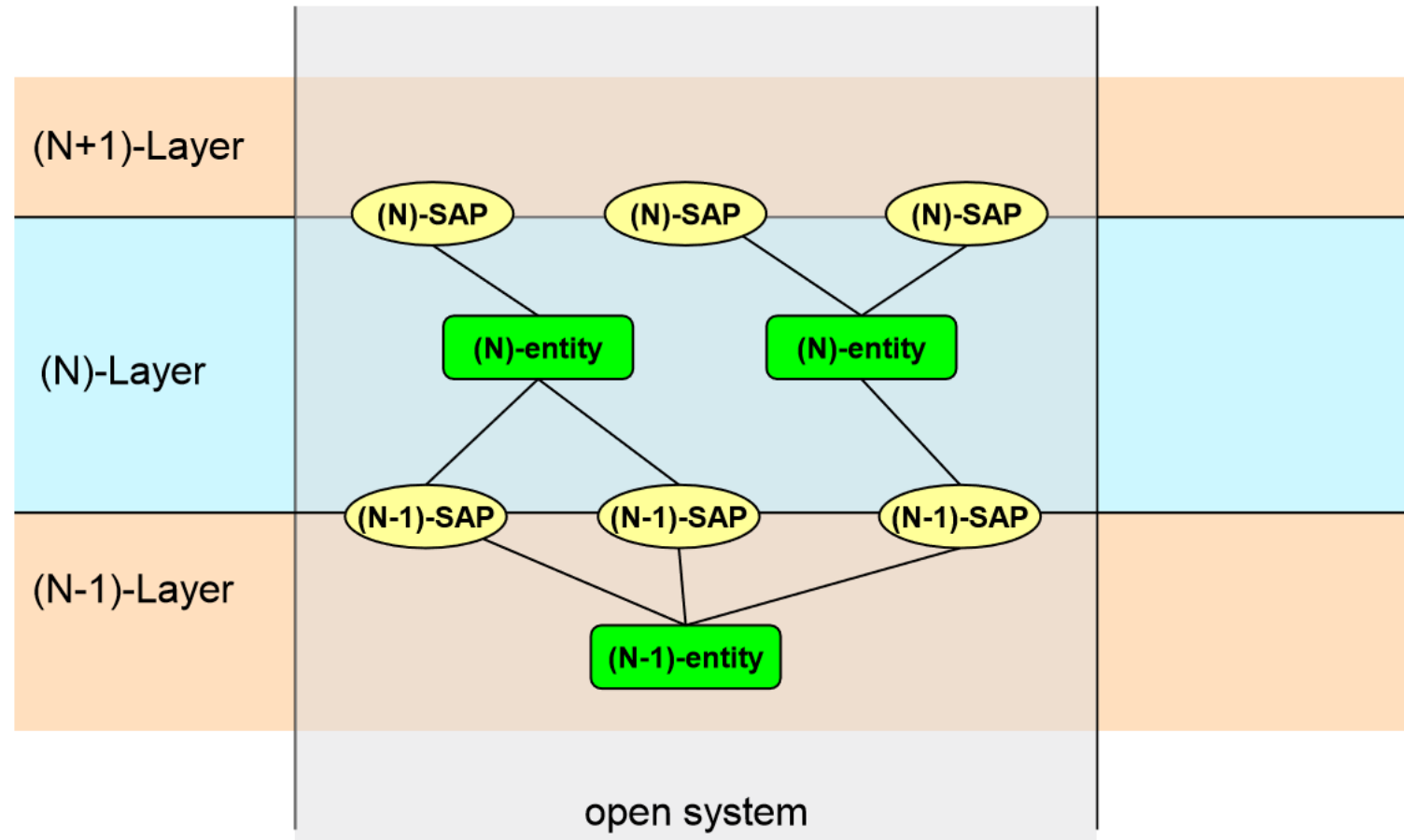
Encapsulation in the TCP/IP Suite



Addressing in the TCP/IP protocol suite

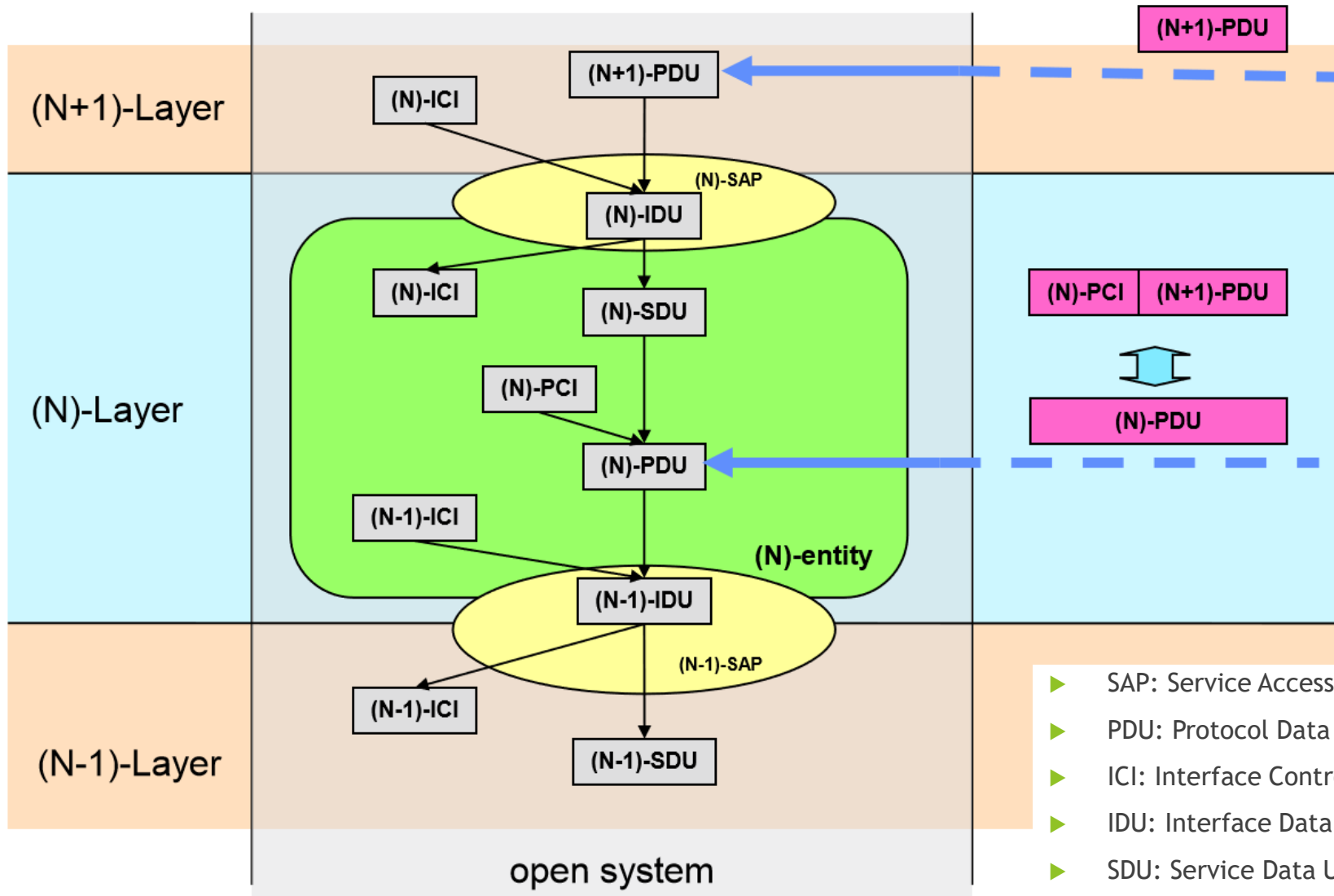
Packet names	Layers	Addresses
Message	Application layer	Names
Segment / User datagram	Transport layer	Port numbers
Datagram	Network layer	Logical addresses
Frame	Data-link layer	Link-layer addresses
Bits	Physical layer	

Layering Principles (1)



► SAP: Service Access Point

Layering Principles (2)



- ▶ SAP: Service Access Point
- ▶ PDU: Protocol Data Unit
- ▶ ICI: Interface Control Information
- ▶ IDU: Interface Data Unit
- ▶ SDU: Service Data Unit
- ▶ PCI: Protocol Control Information

Services and Protocols in the OSI Model

► Services

- In OSI model, each layer provide services to layer above, and 'consumes' services provided by layer below.
- Active elements in a layer called *entities*.
- Entities in same layer in different machines called *peer entities*.

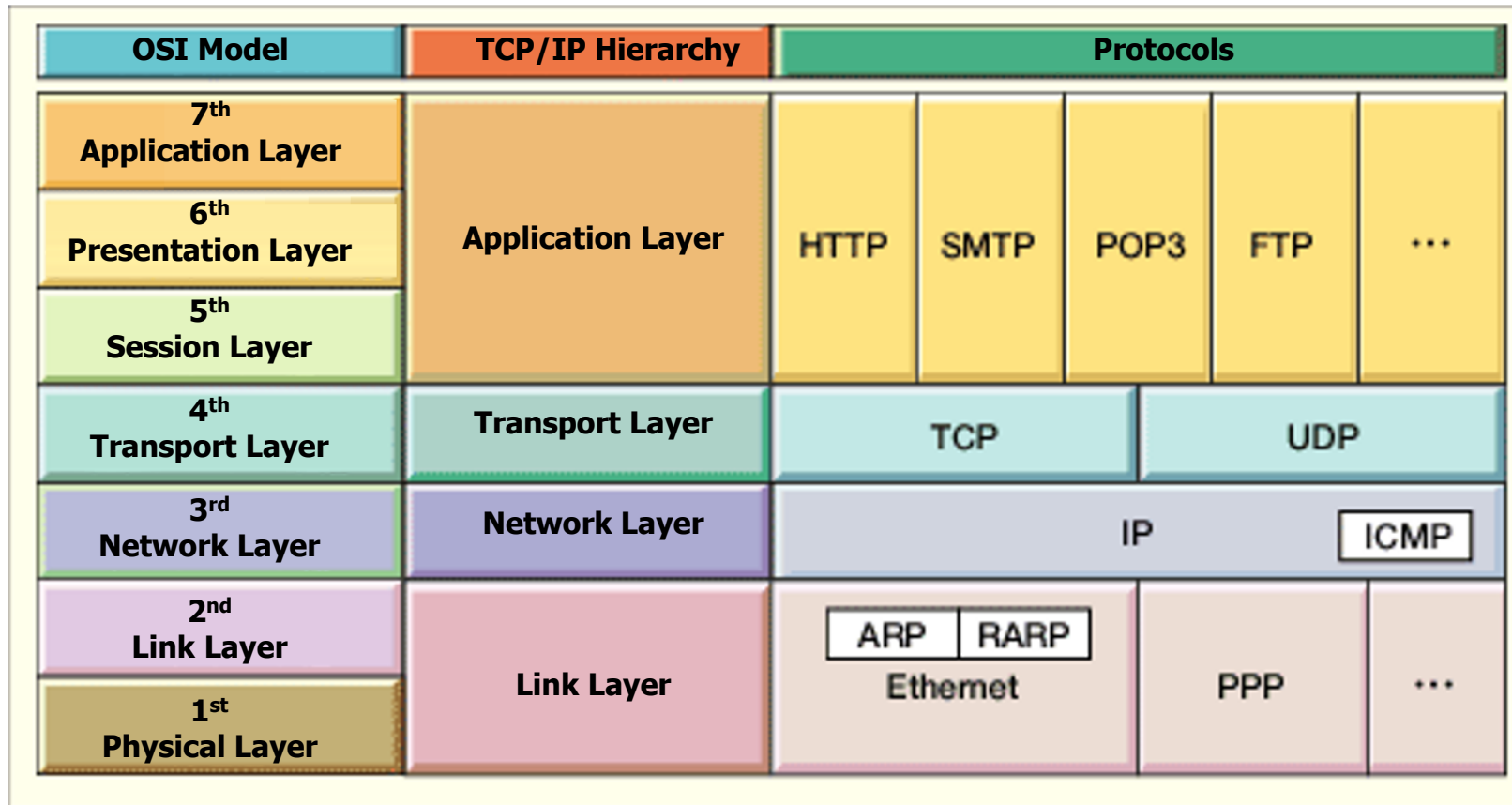
► Protocols

- In diplomatic circles, a protocol is the set of rules governing a conversation between people
- A network protocol is the set of rules governing a conversation between a client and a server
- -> **Syntax, Semantic and Timing**
- A standard protocol is described in RFCs (Requests for Comments)

Service vs. Protocol

- ▶ Service = set of primitives provided by one layer to layer above.
- ▶ Service defines what layer can do (but not how it does it).
- ▶ Protocol = set of rules governing data communication between peer entities, i.e. format and meaning of frames/packets.
- ▶ Service/protocol decoupling very important.

TCP/IP Protocol Stack



- ▶ A protocol stack is a group of protocols that are running concurrently and are employed for the implementation of network protocol suite
- ▶ Every operating system has a protocol stack built in.
- ▶ The protocol stack used on the internet is TCP/IP.

Link Layer : includes device driver and network interface card
Network Layer : handles the movement of packets, i.e. routing
Transport Layer : provides a reliable flow of data between two hosts
Application Layer : handles the details of the particular application

Some Popular TCP/IP Protocols

